

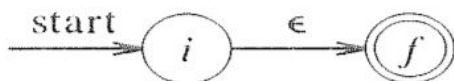
Lecture 8.2: RE to NFA (Thompson's Construction)*Presenter: Warida Rashid**Scribe: Warida Rashid*

Thompson's construction algorithm, also called the McNaughton-Yamada-Thompson algorithm, is a method of transforming a regular expression into an equivalent nondeterministic finite automaton (NFA). Regular Expressions define the pattern for a language and Finite Automaton is suited for implementing the pattern in terms of programming. Therefore, we often have to convert a Regular Expression into a Finite Automaton.

Thompson's Construction Algorithm

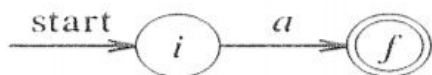
The algorithm works recursively by splitting an expression into its constituent subexpressions. The rules for constructing an NFA consist of some basis for handling the subexpressions with no operators, and inductive rules for constructing larger NFA's from the NFA's for the immediate subexpressions of a given expression.

Basis: For the Regular Expression $R = \epsilon$, the NFA is:



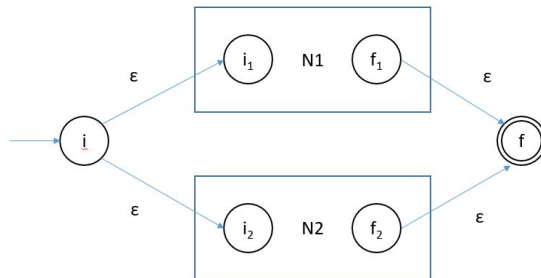
Here, i is the start state and f is the final state.

For a subexpression that is a symbol in the alphabet such as $R = a$, $a \in \Sigma$, the NFA is:

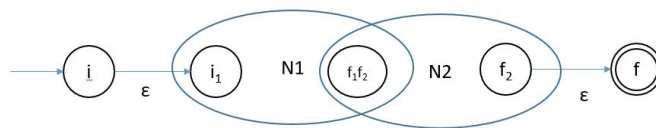


Induction: Suppose $N1$ and $N2$ are the NFAs for the Regular Expressions $R1$ and $R2$ respectively.

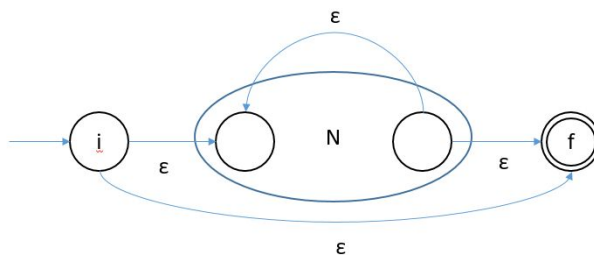
1. The NFA for the RE $R1|R2$ is



2. The NFA for the RE $R1.R2$ is



3. If R is a Regular Expression and N is the NFA for R , then the NFA for the R^* is



Conditions

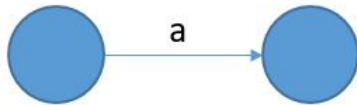
The NFA derived from Thompson's Construction must follow some conditions:

1. There can be only one start state.
2. The start state cannot be accessible from any states i.e. it cannot have transitions to it from other states.
3. There can be only one final state. The final state cannot have any transition from it to other states.
4. The number of transitions leaving any state is at most two.

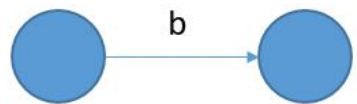
Example:

Consider the Regular Expression **(a|b)*abb**. The steps following the Thompson's Construction algorithm are:

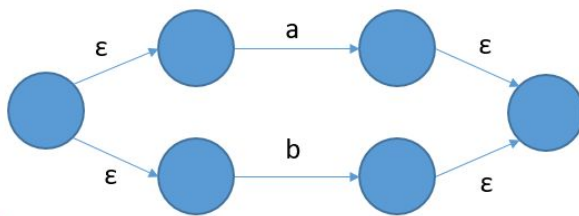
1. Subexpression: a



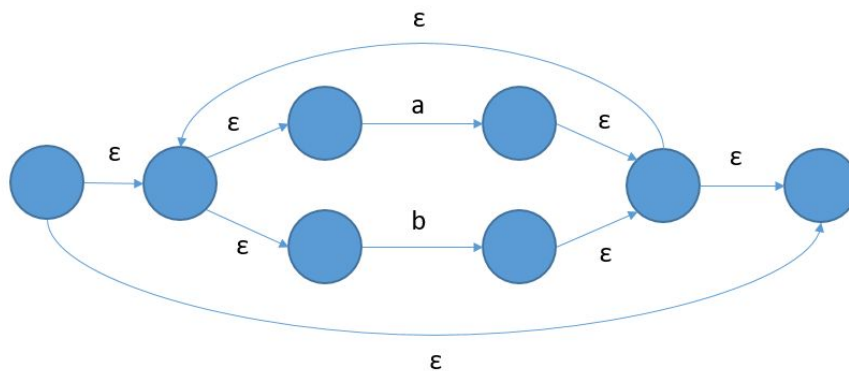
2. Subexpression: b



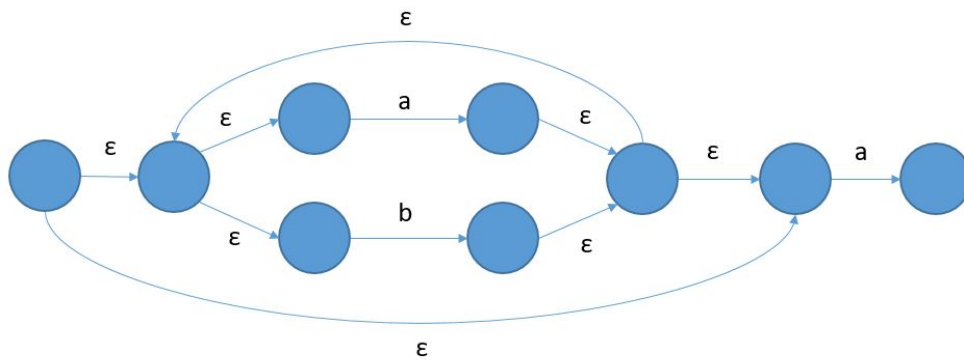
3. Subexpression: a | b



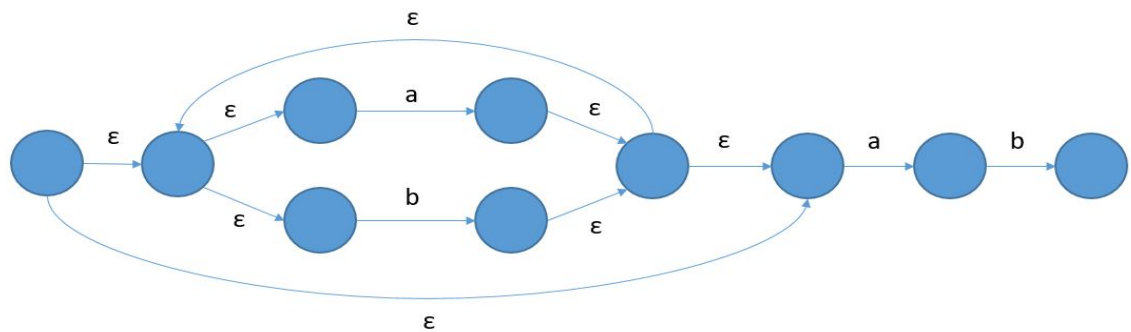
4. Subexpression: (a | b)*



5. Subexpression: $(a \mid b)^*a$



6. Subexpression: $(a \mid b)^*ab$



7. The NFA for the entire expression: $(a \mid b)^*abb$

